

COMPASS Initial Review and Alignment Memo

Name note: Authored by **COMPASS** (agent renamed 2026-06-11; formerly “Master Planner”). See [COMPASS_AGENT_IDENTITY.md](#).

Date: 2026-06-11

Author: COMPASS (Cursor agent)

Purpose: Structured situational review after ingesting v6 guidance (VECTOR handoff), three domain agent reports, theory/manuscript corpus, dissertation proposal, Dakota v03 brief, and advisor memos.

Status: Near-term plan **largely locked** after Charles clarifications (2026-06-11). Advisor update: [20260611_Brief_for_Alex_Gates_brief.md](#) / full.

1. Executive synthesis

Charles Levine's dissertation program has **morphed substantially** since the August 2024 proposal (*Deciphering the Talent Mosaic* — prestige networks, unit eigenvector centrality, three Army-only chapters) while retaining core DNA: **Army officer careers as a closed talent system**, **peer and evaluator context matter**, **network-science and survival methods**, and **Barabási-style cross-domain universality ambition**.

The **current scientific center of gravity** (June 2026) is narrower and more defensible:

***Advancement under constrained distinction** — individuals compete in **local comparison pools** while **global distinction is finite**. Empirically, **advancement probability rises with LOO peer-pool quality through the middle of the distribution, then falls at the elite tier** (inverted-U). This pattern is **established in Army, replicated in NCAA basketball → NBA draft, and preliminary in R1 CS tenure**.*

The project is transitioning from **empirical discovery** to **explanatory modeling** (Wang-style: stylized fact → minimal mechanism → non-obvious predictions). The **immediate convergence risk** is not lack of ideas — it is **parallel expansion** (525/UIC, network extensions, generative axis mismatches, tenure scrape expansion) displacing **minimal model closure**, **prediction list discipline**, and **manuscript draft**.

VECTOR's handoff is operationalized through [COMPASS_Initial_Guidance_v6.md](#) plus the theory corpus ([Vector_Master_Theory_and_Modeling_Notes_4.md](#), [Tier1_Narrative_Outline.md](#), Dakota v03 RTF, Wang model notes). There is no separate [VECTOR_report_to_COMPASS.md](#); v6 + 5-Manuscript folder is the VECTOR lane per Charles.

2. Established vs preliminary — findings by domain

Setting	Agent	Outcome	Pool construct	Inverted-U	Cox / survival	Manuscript-ready?
1 — Army	CODA	CPT→MAJ promotion vs attrition	Senior-rater LOO pool TB (<code>pool</code> minus <code>mean</code>)	Established — CIF Q-bins + Cox quadratics	Yes — Cell 11 CIF + Cell 12 cause-specific Cox	Strongest — figures + HR tables exist; estimand language needs care
2 — Basketball	SCOUT	NBA draft (<code>Y_draft</code>)	LOO teammate <code>poolq_loo</code>	Replicated — ventile/binning draft rate	Partial — Wang ladder (bins→LPM→logit); not time-to-event CR	Strong empirical; generative claims must be qualified
3 — Academia	PEER	Tenure (Asst→Assoc) vs attrition	LOO dept <code>poolq_loo_mean</code>	Preliminary — stage 9, 18 bins, elite bin dip	 Planned (Layer B) — <code>540</code> ends at Cell 9; port <code>520</code> → Cells 10/10.5/12	Preliminary figure + honest limitations only

Confidence tags (COMPASS)

- **High confidence:** Army and basketball show the same *qualitative* inverted-U on LOO peer-quality axes (descriptive / binned / CIF layers).
- **Medium confidence:** Cox quadratic + interaction terms in Army support curvature on hazard scale; basketball LPM/logit yields turning point L^* .
- **Low-medium confidence:** Tenure stage-9 pattern (small N per bin, ~58% censoring, noisy OpenAlex linkage, unconditional bins).
- **Not yet supported:** Claim that **one minimal generative score** reproduces empirical inverted-U on L_Q **LOO axis** in simulation (SCOUT June finding: inverted-U on `team_mean`, mostly decreasing on L_Q).

3. Theory / minimal model state (VECTOR framing)

Working thesis (cross-domain)

From `Vector_Master_Theory_and_Modeling_Notes_4.md` and `Tier1_Narrative_Outline.md`:

Advancement depends on the interaction between **local peer quality** and **scarce distinction**. Stronger pools improve development/signaling up to a point; at the elite tier, **congestion, signal compression, and finite slots** dominate → inverted-U.

Tier 1 minimal objects

Object	Meaning	Empirical proxy (basketball)
Y	Global advancement (distinction received)	<code>Y_draft</code>
L_{ijt}	Amalgamated local environment	Rolled into Q first
Q_{ijt}	LOO peer quality	<code>poolq_loo</code>
Λ_t	Global distinction capacity	Draft slots / promotion boards / tenure lines
A_{ijt}	Own ability (control, not headline)	PPM z

Minimal generative score (Alex / SCOUT):

$$S_i = A_i - \lambda \cdot L_{C,i}$$

where L_C = LOO viable-peer congestion. This is the **domain-agnostic minimal claim** — not the 539 notebook bundle.

Theory vs model discipline (Alex, Apr 30 2026)

`2026_0430_Paper7_feedback.md` and `2026_0507_Alex_Gates_Post_Meeting_Simulation_Memo.md` sharpen:

- **Theory** may include development, evaluator compression, sorting, prestige, networks.
- **Minimal model** should formalize the **smallest** set that reproduces the nonlinear pattern.
- **Quadratic in Q** is an **empirical diagnostic**, not the generative first principle.
- **Assortative pool formation + local comparison** is the generative story Alex emphasized (PD10–PD12).

Gaps vs empirical evidence

Gap	Severity	Notes
Generative inverted-U on same axis as 530 (poolq_loo)	High	Same score, different Plot B axis → different shape (June 2026)
PEER Layer B Cox not archived	Medium for draft / High for submission	Soft gate: CELL 9 + limitations OK for VECTOR draft now; basic Cell 12 before submission; not urgent during draft phase (Charles 2026-06-11)
Mechanism decomposition (B vs D) not separately identified	Medium	Tier 1 intentionally rolled-up; predictions list underdeveloped
Evans structural misclassification integrated in theory, not in minimal sim	Medium	Manuscript optional for v1
Network-science extensions (exposure vs comparison nets)	Low for v1	Dakota §8 + Barabási brief — explicitly defer

4. Cross-agent tensions and over-claim risks

4.1 Estimand conflation (highest manuscript risk)

Tension	Agents involved	Planner ruling
CIF bar panels vs Cox HRs	CODA	Keep layers distinct in prose; CIF = binned cumulative incidence; Cox = cause-specific hazards with censoring at competing event
Cell 11 CIF estimator simplicity	CODA	Do not claim full Aalen–Johansen; last-snapshot bin assignment is implementation fact
Plot B axis: <code>team_mean</code> vs L_Q	SCOUT / VECTOR	Not interchangeable ; empirical paper leads with L_Q ; generative POC may use <code>team_mean</code> — must say so explicitly
539 vs 538D relationship	SCOUT	Parallel bundles, not parent/child; minimal claim = equation , not notebook
Fine–Gray mentioned in Dakota/PEER	PEER / VECTOR	Code uses cause-specific Cox + empirical CIF; verify before manuscript claims Fine–Gray
“168 schools” inference-ready	PEER	168 <code>uni_slug</code> ≠ 168 equal-quality departments; distinguish coverage vs inference sample

4.2 Duplicated effort

Duplication	Recommendation
Multiple Alex briefing outlines (v1–v4)	Canonical: <code>alex_gates_briefing_structure_outline_v4.md</code>
Multiple Scout status docs	Canonical: <code>Scout_Status_Update_for_VECTOR_Laszlo_Briefing_2026-06-02.md</code>
<code>Vector_to_Scout_Tier1_Modeling_Direction.md</code> vs Alex spine	Not sole canonical per CODA Q11; use Alex spine + June Scout status
535 vs 538 empirical paths	Consolidate on 538/538D unless Charles revives 535
537 generative sim	Frozen — benchmark only

4.3 Hidden assumptions carrying weight

- 1. **LOO peer quality as sufficient statistic for local environment** — rolls development + congestion into one proxy (Tier 1 design choice).
- 2. **Three settings in one manuscript** — Charles confirmed 2026-06-08 (CODA report §10); dissertation bar = all three.
- 3. **Army proprietary data** — basketball/tenure are public-facing legs; Army figures may be illustrative without micro-data release.
- 4. **PPM+z as cross-domain performance harmonizer** — SCOUT still flags BPM/OBPM as open.
- 5. **Transcripts folder** — referenced in v6 but **not present in repo workspace**; advisor signal absorbed via memos (`2026_0507` , `2026_0430` , `Alex_Tier1_Sequential_Model_Outline` , `TENURE_DATA_GAMEPLAN` § advisor).

4.4 Sequencing tensions — resolved vs open

#	Issue	Status (2026-06-11)
1	Cell 12 Cox on tenure before Setting 3 prose	Resolved — soft gate: CELL 9 + limitations OK for VECTOR draft now; Cell 12 not urgent during draft; basic Cox before submission; Fine–Gray deferred
2	SCOUT generative priority	Resolved — manuscript-first (Path A / Path II): honest layering + axis table; LOO generative match deferred; nesting prose blocking
3	CODA parallel track (525/TB-stratify vs pool audit vs manuscript)	Open — Charles has not ranked; default defer per near-term plan
4	OpenAlex confidence filter for tenure inference sample	Open — HIGH only vs HIGH+MEDIUM vs MULTI with disclaimer

5. Advisor alignment sketch (Alex Gates — preliminary)

Full `MENTAL_MODEL_OF_ALEX.md` deferred to Phase 1 embed in near-term plan; preliminary read:

What problem Alex believes we are solving

Show that *nonlinear advancement* emerges when *talent sorts into local pools* and *selection uses local comparison*, not raw ability alone — with a *minimal generative story*, not a quadratic baked in.

What evidence would convince Alex

- Cross-domain **empirical replication** (done for 2.5/3 settings).
- **Generative check**: ability-only fails; adding congestion/local comparison bends the curve.
- **Honest axis/estimand language** — no over-claim that one notebook “decomposes” another.
- **Wang ladder**: bins → transparent quadratic as **diagnostic** → logit L^* → generative mechanism.

What Alex would likely cut

- Network-science extensions before minimal model + predictions.
- Copying observed team means into generative assignment as “the model.”
- Presenting quadratic as **the** mechanism.
- Chasing perfect tenure scrape before end-to-end measurement on existing data.

What Alex would likely prioritize

- **PD11 threads:** (C) promotion score vs local rank tweaks → (A) soft-assignment pool formation → (B) mean × dispersion EDA.
- **End-to-end curves** with sample-loss accounting (PEER delivered this ethos).
- **Non-obvious predictions** (heterogeneity in draft-relevant tail; threshold congestion).

Risks Alex appears focused on

- Methodological overreach (causal claims, Fine–Gray without implementation).
- Generative/empirical axis mismatch presented as replication.
- Distraction from **paper** by 525 coding or URL scraping.

Laszlo Barabási (strategic): empirical replication + minimal mechanism + network-science **future** hooks
([barabasi_briefing_outline_v2.md](#)). Lead with stylized fact, not closed proof.

6. Dissertation evolution — original vs current

Dimension	Aug 2024 proposal	June 2026 program
Title/framing	Talent mosaic, unit prestige, OCN eigenvector centrality	Advancement under constrained distinction
Domains	Army only (3 chapters)	Army + basketball + academia
Core RQs	Prestige units, peer effects, TB tradeoffs	Inverted-U + minimal mechanism + predictions
Methods	OCN/OPN networks, survival	Survival + LOO pool metrics + Tier 1 generative lab
Committee	Barabási chair, Gates, Vespignani, Dakota Murray	Same; Dakota received v03 brief

Retained: Army as anchor; Evans misclassification; competing risks; people-analytics audience; Wang/Barabási scientific style.

Deprioritized for immediate manuscript: Full OCN prestige validation (H1a–c), 525 UIC consistency as *paper* centerpiece (still valuable for Army depth / dissertation later).

7. Charles decisions (locked) and remaining open questions

Locked (COMPASS thread, 2026-06-11)

#	Topic	Decision
1	Hard deadline	Summer–Fall 2026 core manuscript draft/submission
2	Tenure Cox gate	Soft gate — see §4.4 #1
3	SCOUT generative priority	Manuscript-first — see §4.4 #2
—	Terminology	LOO in any name for quantities excluding self
—	Alex comms	Plan update briefs (<code>*_Brief_for_Alex_Gates_*.md</code>) — Charles’s calls, not a decision questionnaire

Still open

- 4. CODA parallel track (§4.4 #3) — TB-stratify on AWS? pool-size audit before manuscript?
- 5. Transcripts location — `transcripts/*_Paper_directions*` not in workspace; memos sufficient?
- 6. Army data in planning docs — what can be named externally?
- 7. Stale file flags — any v6 paths superseded beyond agent reports?
- 8. Who owns Dakota RTF refresh when domain status shifts?
- 9. `PERF_METRIC` lock (PPM+z) vs BPM for cross-domain comparability?
- 10. OpenAlex confidence policy for tenure (§4.4 #4).

8. Draft agent question queue

File	Rationale
<code>20260611_1626_COMPASS_to_VECTOR_questions.md</code>	Minimal model closure criteria; prediction prioritization; manuscript section ownership
<code>20260611_1626_COMPASS_to_SCOUT_questions.md</code>	Generative L_Q gap vs manuscript scope; axis figure requirements
<code>20260611_1626_COMPASS_to_PEER_questions.md</code>	Setting 3 maturity target; Cox timeline; inference sample definition
<code>20260611_1626_COMPASS_to_CODA_questions.md</code>	Army estimand language for manuscript; deferred TB-stratify / pool audit

Charles routes these to agents; COMPASS will not assume answers.

9. Provisional answer — shortest defensible path

If the objective is to complete the minimal model, generate robust predictions, and produce a publishable manuscript, what is the most scientifically defensible path from today?

Locked path (Charles, 2026-06-11):

```
LOCK CLAIM LANGUAGE (all agents)
↓
SCOUT: Manuscript exports + axis table + nesting chain for VECTOR §3
      (defer L00 generative bin-for-bin match; parallel north-star sprint if time)
↓
VECTOR: Draft Wang-structure manuscript – empirical triad → minimal model → predictions
↓
PEER: Layer B Cell 12 Cox before submission only (not urgent during draft; CELL 9 OK for VECTOR now)
↓
CODA: Manuscript figures + estimand-checked captions; 525/TB-stratify deferred
↓
PREDICTIONS: Lock 2 testable (heterogeneity + one dispersion/stratification candidate)
↓
SUBMIT Summer–Fall 2026; defer networks, Fine–Gray, full L00 generative match
```

Detail: `PROJECT_STATUS_AND_NEAR_TERM_PLAN.md` and `20260611-Brief_for_Alex_Gates_full.md`.

10. Review status

- §§1–4 **situational understanding**: Charles affirmed via clarification thread (2026-06-11).
- §9 **path**: Locked for near-term; open items in §7.4–10 remain.
- `PROJECT_STATUS_AND_NEAR_TERM_PLAN.md`: Active living doc; revise when agent answers land.
- Alex: `20260611-Brief_for_Alex_Gates_brief.md` / `_full.md` — plan update (Charles’s calls), not a decision memo.

End initial review memo.